TASFIA JANNAT TASFI

Human–Computer Interaction | Human-Centered AI | NLP

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Research Profile

I am a Computer Science and Engineering graduate interested in understanding how people interact with AI-enabled systems and how interface design influences their decisions. My undergraduate thesis combines web-based interface design, NLP, and a controlled user study to examine how different explanations affect people's understanding and use of AI outputs. I enjoy building interactive research prototypes and evaluating them with users, and I hope to continue studying how AI system design shapes people's choices, understanding, and sense of control.

Research Interests: Human-Centered AI; Responsible AI; Human–AI Interaction; User Agency; Social Computing; Explainability; Appropriate Reliance; NLP

Education

B.Sc. in Computer Science and Engineering **11 Aug. 2026**
Rajshahi University of Engineering & Technology (RUET) *Rajshahi, Bangladesh*

CGPA: 3.77/4.00 | **Program:** BAETE-accredited CSE program

Research Experience

Undergraduate Thesis: Designing and Evaluating Explanation Interfaces for AI-Assisted Medical Misinformation Detection
Human–Computer Interaction | NLP | Explainable AI | Human–AI Decision Making

- Designed and implemented **four web-based explanation interfaces** that communicate AI assessments, linguistic indicators, calibrated probabilities, uncertainty, evidence, and sources; built the end-to-end research prototype with **Next.js and FastAPI**.
- Conducted a controlled HCI study with **60 participants and 1,200 judgment trials**. The most informative interface improved judgment accuracy from **73.7% to 90.0%** and reduced over-reliance on incorrect AI advice from **77.8% to 37.8%**.
- Developed the NLP backend using **RoBERTa, calibrated uncertainty, and conformal selective prediction** on a multi-source corpus of **17,611 samples**. The selective system automatically handled **91.56%** of cases with macro-F1 **0.9806**, while referring uncertain cases for human verification.

Publications

Jabunnesa Jahan Sara, Shahriar Rizvi, and Tasfia Jannat Tasfi. *Beyond Artifacts: Semantic Deepfake Detection and Digital Provenance Tracking with Multimodal Large Language Models.* 2025 28th International Conference on Computer and Information Technology (ICCIT), 2025.
DOI: https://doi.org/10.1109/ICCIT68739.2025.11490141

Developed a multimodal pipeline for semantic deepfake analysis and web-based provenance tracing, moving beyond purely visual artifact detection.

Nahid Montasir Rifat, Tasfia Jannat Tasfi, Foyez Ahmed Dewan, and Barshon Sen. *Evaluating Creative Reasoning and Lateral Thinking in Instruction-Tuned LLMs for Bengali.* 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), 2026.
DOI: https://doi.org/10.1109/QPAIN69676.2026.11546604

Built a parallel Bangla–English benchmark for instruction-tuned LLMs and analyzed cross-lingual failure modes such as mistranslation, cultural mismatch, and logical inconsistency.

Selected Achievements

- **1st Place, UIHP 5th Cohort (WayMakers)** — recognized for startup innovation, market validation, and MVP development.
- **Champion**, Huawei Seeds for the Future Tech-Business Competition, 2024.
- **Champion**, Hult Prize at RUET — Startup Innovation, 2023.
- **Government General Scholarship** — awarded for all four years of undergraduate study.
- **Finalist**, International Youth Math Challenge — Top 20% globally, 2024.
- **70% Merit Scholarship**, ISCEA PTAK Prize, 2024.
- **Government Talent Pool Scholarship** — awarded for academic excellence in high school.

Selected Systems & Design Projects

EarlyMind — AI-Assisted Child Development Interface. Built an interactive web interface for child-development assessment activities, with adaptive dashboards and progress feedback informed by cognitive-load considerations.
Project: github.com/Shion115/EarlyMind

BongoSeek — LLM-Powered Conversational Search. Built a conversational search system with context-aware query handling and dialogue memory for coherent multi-turn interaction.
Project: github.com/Shion115/Bongo_Seek

Professional & Leadership Experience

Co-Founder, Delibro **Nov. 2024 – Present**
Social Computing Platform for Community-Based Delivery
- Co-founded a community-oriented delivery platform centered on trust, reciprocity, safety, and human connection; contributed to interface design, prototyping, product development, and user-oriented validation. The project received support through the **World Bank-supported University Innovation Hub**.

Official Ambassador **Jun. 2025 – Present**
International Computer Science Competition
- Organized programming outreach, mentorship activities, and student community initiatives. **Profile:** icscompetition.org/en/ttasfi

Industrial Trainee **Mar. 2025**
BRAC IT
- Worked on front-end interface design and web-development workflows.

App Developer **Jun. 2022 – Jan. 2023**
bdapps
- Developed and deployed PHP-based applications involving interface design, backend integration, and deployment.

Research & Technical Skills

HCI & Research	Human–AI Interaction, User Studies, Interface Evaluation, UX Design, Prototyping
Methods & Analysis	Controlled Experiments, Experimental Evaluation, Uncertainty Calibration, Conformal Prediction, Classification Metrics
Web & Prototyping	Next.js, JavaScript, HTML/CSS, FastAPI, Flask, Figma
AI / NLP	Python, Transformers, RoBERTa, Hugging Face Transformers, scikit-learn
Programming & Tools	Python, C++, JavaScript, PHP, Git, Jupyter Notebook, Google Colab, pandas

References

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